

Comparative Performance Report for cbt-260420-o01-directreads-3way-unbalanced vs cbt-260421-o01-directreads-3way-balanced

Table of contents

- Comparison summary for cbt-260420-o01-directreads-3way-unbalanced vs cbt-260421-o01-directreads-3way-balanced
- Response Curves
 - Sequential Read
 - Random Read
 - Random Read/Write
- Configuration yaml files
 - results

Comparison summary for cbt-260420-o01-directreads-3way-unbalanced vs cbt-260421-o01-directreads-3way-balanced

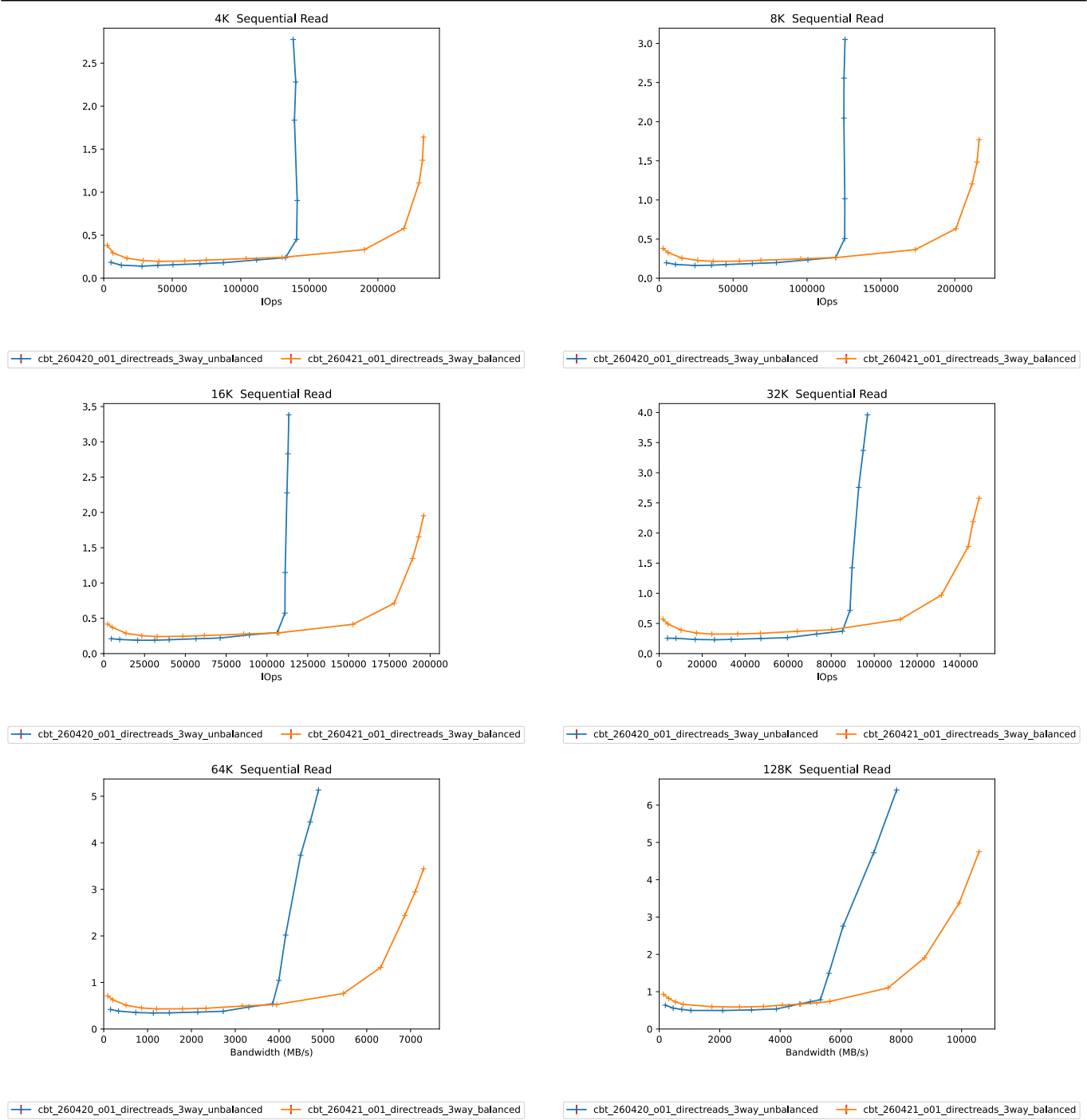
Sequential Read	cbt_260420_o01_directreads-3way-unbalanced	cbt_260421_o01_directreads-3way-balanced	%change throughput	%change latency
4K	141151 IOps@0.9ms	233361 IOps@1.6ms	65%	78%
8K	125729 IOps@3.1ms	216424 IOps@1.8ms	72%	-42%
16K	113416 IOps@3.4ms	195992 IOps@2.0ms	73%	-41%
32K	96872 IOps@4.0ms	148786 IOps@2.6ms	54%	-35%
64K	4900 MB/s@5.1ms	7297 MB/s@3.4ms	49%	-33%
128K	7853 MB/s@6.4ms	10579 MB/s@4.8ms	35%	-25%
256K	11510 MB/s@8.7ms	13479 MB/s@6.2ms	17%	-29%
384K	12117 MB/s@16.6ms	13493 MB/s@7.4ms	11%	-55%
512K	12939 MB/s@20.7ms	12882 MB/s@20.8ms	-0%	0%
768K	13065 MB/s@30.8ms	12338 MB/s@8.1ms	-6%	-74%
1024K	13132 MB/s@30.7ms	12442 MB/s@5.4ms	-5%	-82%
2048K	11256 MB/s@11.9ms	12196 MB/s@5.5ms	8%	-54%
4096K	10452 MB/s@12.8ms	11492 MB/s@11.7ms	10%	-9%

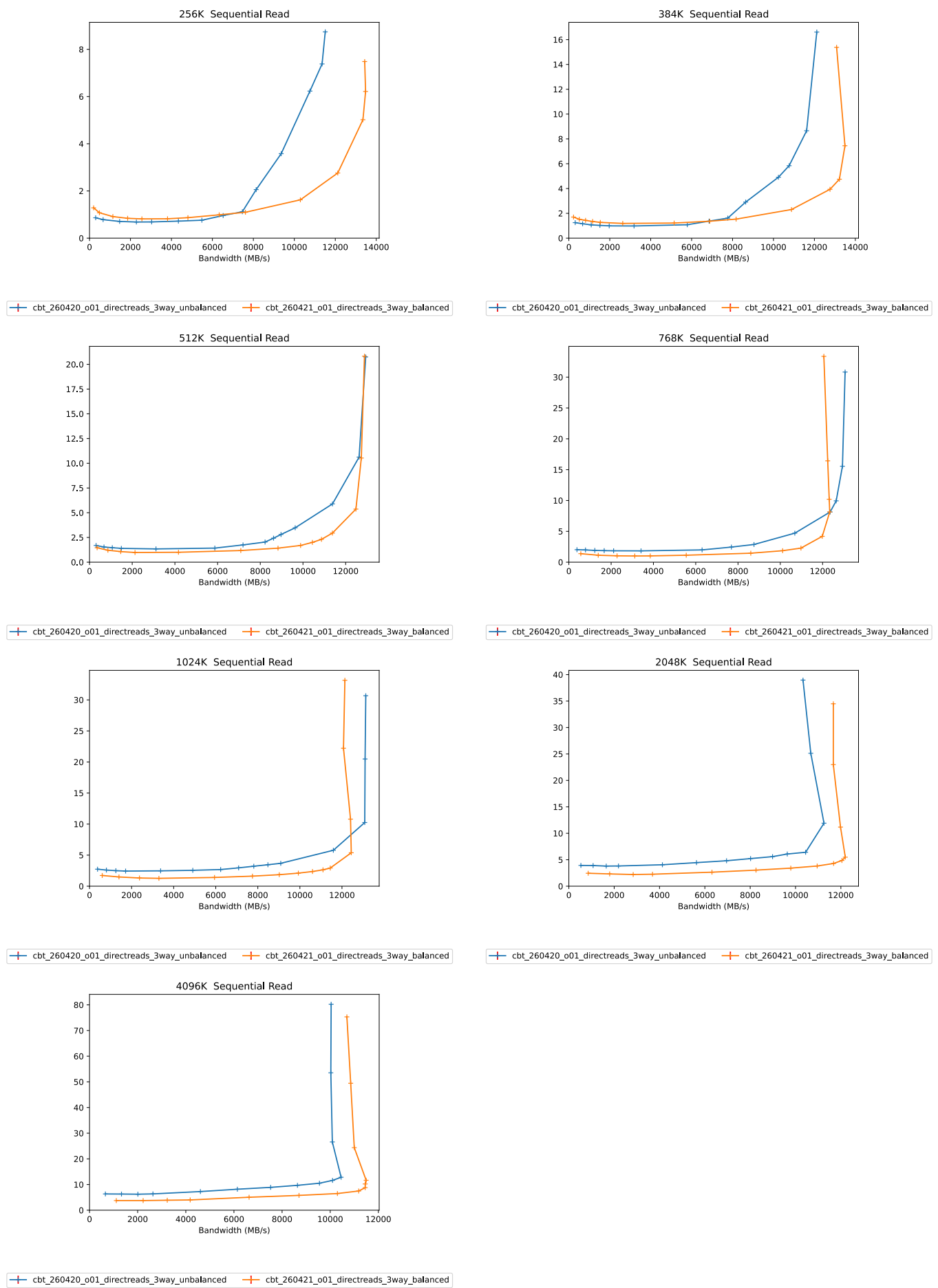
Random Read	cbt_260420_o01_directreads-3way-unbalanced	cbt_260421_o01_directreads-3way-balanced	%change throughput	%change latency
4K	162718 IOps@2.4ms	197591 IOps@1.6ms	21%	-33%
8K	152979 IOps@2.5ms	190066 IOps@1.7ms	24%	-32%
16K	143213 IOps@1.8ms	177373 IOps@1.8ms	24%	0%
32K	121288 IOps@2.1ms	151755 IOps@2.1ms	25%	0%
64K	6209 MB/s@3.4ms	7787 MB/s@3.2ms	25%	-6%
128K	9484 MB/s@5.3ms	10944 MB/s@3.1ms	15%	-42%
256K	12240 MB/s@6.8ms	12862 MB/s@2.6ms	5%	-62%
384K	12638 MB/s@8.0ms	13010 MB/s@3.9ms	3%	-51%
512K	13077 MB/s@5.1ms	12307 MB/s@2.7ms	-6%	-47%
768K	12654 MB/s@7.9ms	12185 MB/s@3.1ms	-4%	-61%
1024K	12651 MB/s@5.3ms	12102 MB/s@5.5ms	-4%	4%
2048K	10981 MB/s@12.2ms	12218 MB/s@5.5ms	11%	-55%
4096K	10706 MB/s@12.5ms	11545 MB/s@10.2ms	8%	-18%

Random Read/Write	cbt_260420_o01_directreads-3way-unbalanced	cbt_260421_o01_directreads-3way-balanced	%change throughput	%change latency
4K_70/30	94440 IOps@4.1ms	101705 IOps@3.8ms	8%	-7%
16K_70/30	84205 IOps@3.0ms	92235 IOps@4.2ms	10%	40%
64K_70/30	4034 MB/s@6.2ms	4342 MB/s@5.8ms	8%	-6%
64K_30/70	2891 MB/s@8.7ms	2924 MB/s@8.6ms	1%	-1%

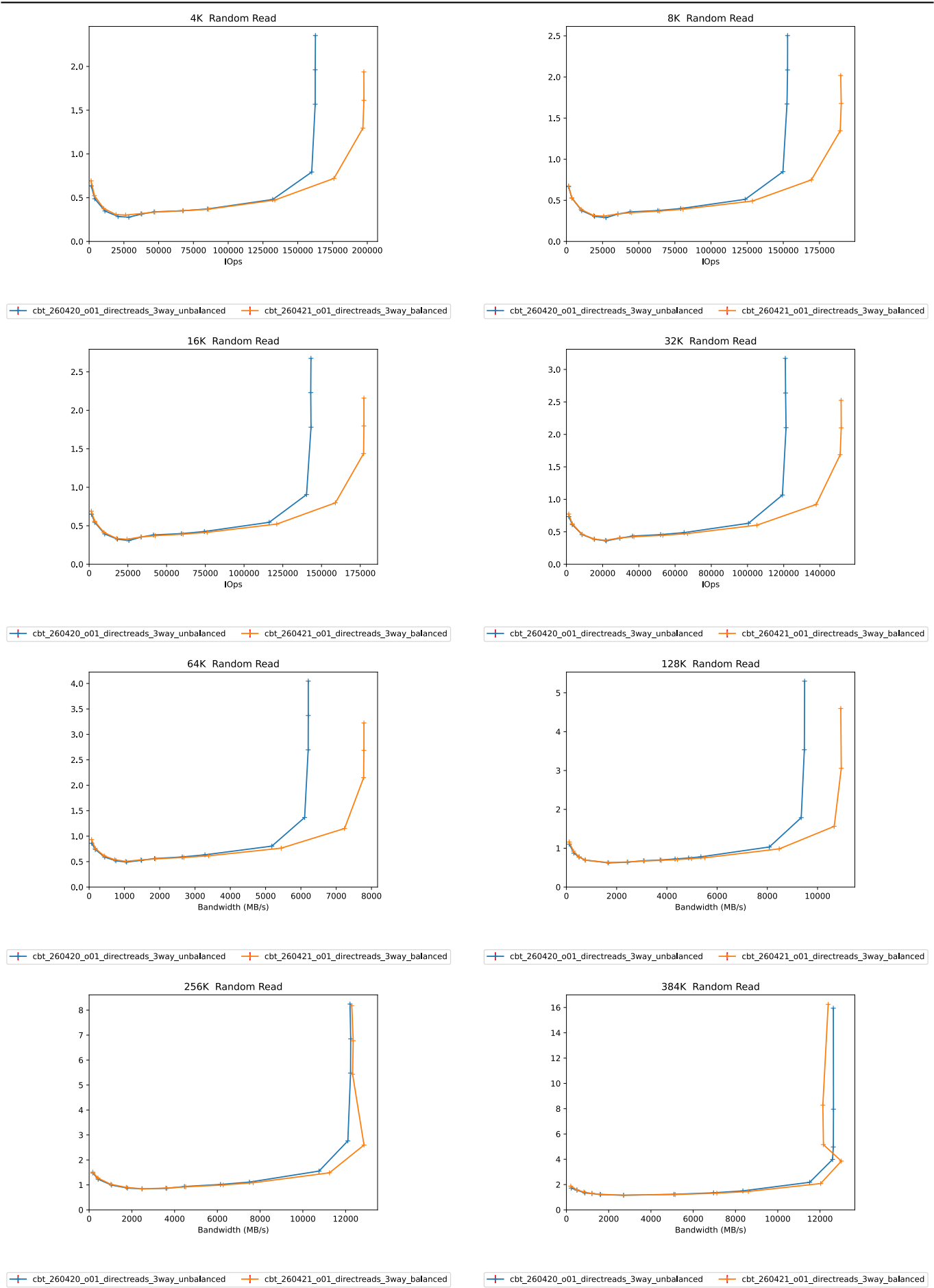
Response Curves

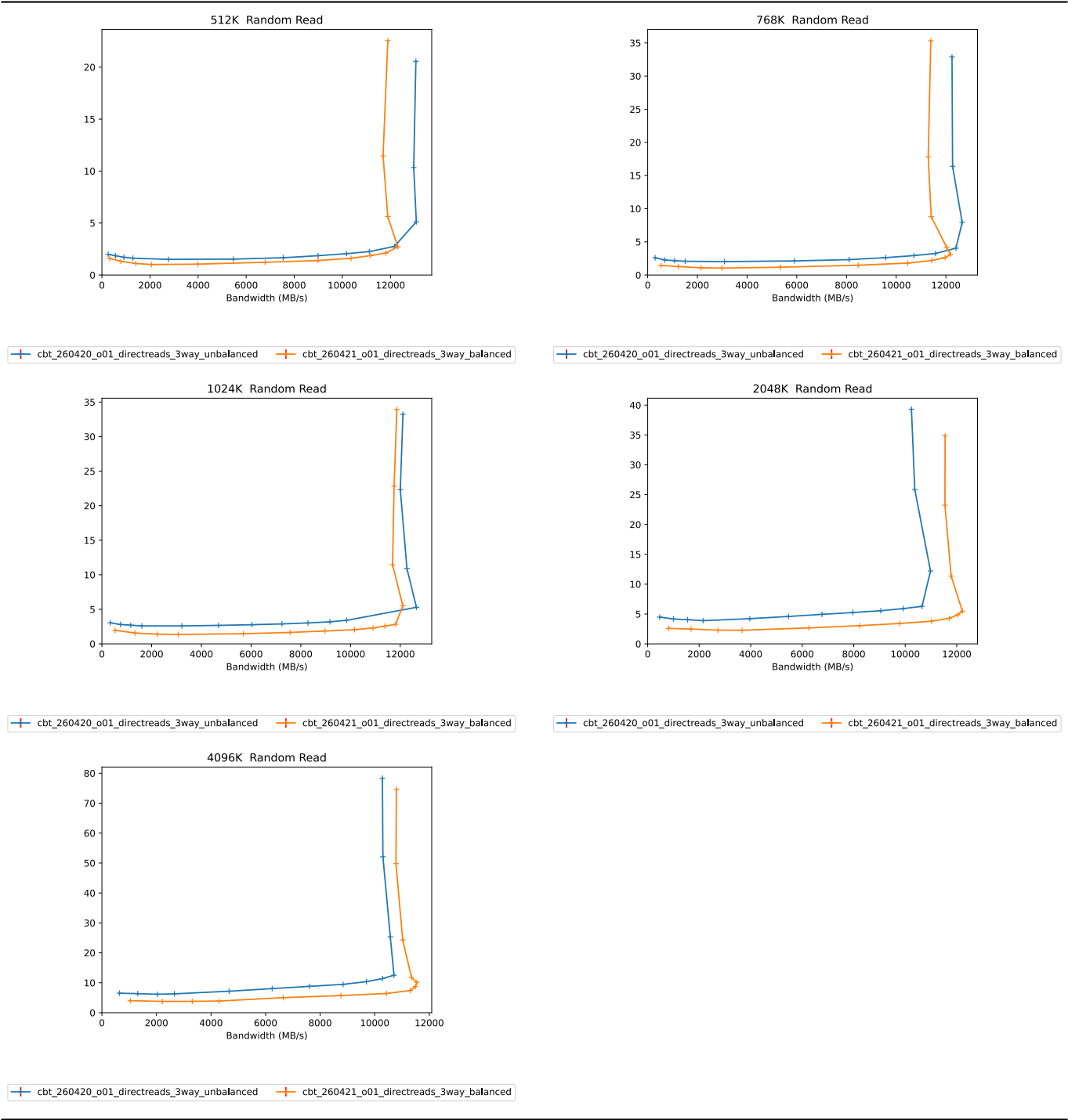
Sequential Read



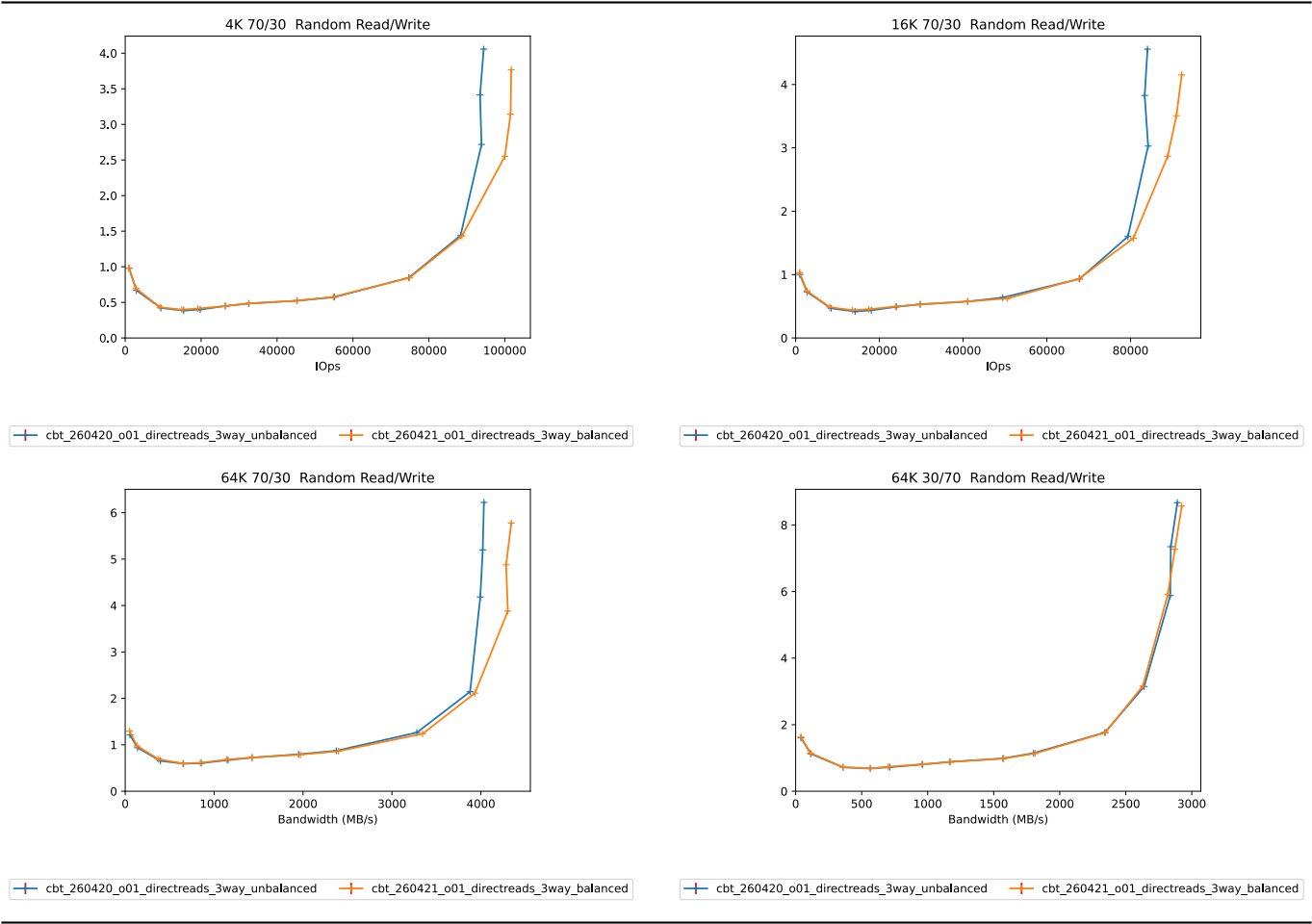


Random Read





Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbdfio:
  cmd_path: /usr/local/bin/fio2
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbdfio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128krandomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128ksequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12

```

```
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
```

```

- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2

```

- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

256ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 262144
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

2Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

2Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1
- 2

```
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
```

```
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
```

```
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
```



```
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

64kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 65536
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

64krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 65536
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

64ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 65536

```
total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 8192
```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null

```

```
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
  - --- server1 ---
osds_per_node: 8
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: ljsanders
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```